



Bansilal Ramnath Agarwal Charitable Trust's
VISHWAKARMA INSTITUTE OF TECHNOLOGY – PUNE
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Bibwewadi and Kondhwa campuses

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Home Assignment 4

Write note and demonstrate using python programming

1

Paired t-test on before & after training results of employees.

2

Independent two-sample t-test comparing two machinery performance.

1. PAIRED t-TEST (Before & After Training)

Objective

To determine whether training has significantly improved employee performance.

Concept

A paired t-test is used when we compare two related observations:

- The same employees are measured twice
- Observations are dependent
- Example: before and after training performance

Hypotheses

- H_0 : There is no difference in performance (mean difference = 0)
- H_1 : There is a significant difference in performance

Sample Data

Before Training: 45, 50, 55, 60, 48, 52, 49

After Training: 50, 55, 60, 65, 52, 58, 54

```
[5]: import numpy as np
      from scipy import stats

      before = np.array([45, 50, 55, 60, 48, 52, 49])
      after = np.array([50, 55, 60, 65, 52, 58, 54])

      d = after - before

      mean_d = np.mean(d)
      std_d = np.std(d, ddof=1)
      n = len(d)

      t_stat = mean_d / (std_d / np.sqrt(n))

      # Degrees of freedom
      df = n - 1

      t_critical = stats.t.ppf(1 - 0.025, df)

      print("t-statistic:", t_stat)
      print("t-critical:", t_critical)

      if abs(t_stat) > t_critical:
          print("Reject H0")
      else:
          print("Fail to reject H0")

      t-statistic: 22.912878474779202
      t-critical: 2.4469118511449692
      Reject H0
```

Interpretation

The calculated t-statistic is greater than the critical value, indicating that training improves employee performance.

2. INDEPENDENT TWO-SAMPLE t-TEST (Machinery Performance)

Objective

To compare the performance of two different machines.

Concept

Used when comparing two independent groups:

- Machine A vs Machine B
- No relation between observations

Hypotheses

- $H_0: \mu_1 = \mu_2$ (Both machines perform equally)
- $H_1: \mu_1 \neq \mu_2$ (Performance differs)

Sample Data

Machine A: 102, 105, 98, 100, 110, 108, 104

Machine B: 95, 97, 92, 96, 94, 93, 91

Interpretation

There is a significant difference in performance between the two machines. Machine A performs better than Machine B.

```
[6]: import numpy as np
from scipy import stats

machine_A = np.array([102, 105, 98, 100, 110, 108, 104])
machine_B = np.array([95, 97, 92, 96, 94, 93, 91])

mean_A, mean_B = np.mean(machine_A), np.mean(machine_B)
std_A, std_B = np.std(machine_A, ddof=1), np.std(machine_B, ddof=1)

n1, n2 = len(machine_A), len(machine_B)

t_stat = (mean_A - mean_B) / np.sqrt((std_A**2 / n1) + (std_B**2 / n2))

# Degrees of freedom
df = n1 + n2 - 2

t_critical = stats.t.ppf(1 - 0.025, df)

print("t-statistic:", t_stat)
print("t-critical:", t_critical)

if abs(t_stat) > t_critical:
    print("Reject H0")
else:
    print("Fail to reject H0")

t-statistic: 5.460620075368335
t-critical: 2.1788128296634177
Reject H0
```

Final Conclusion

1. Paired t-test shows training significantly improves employee performance.
2. Independent t-test shows a significant difference between machinery performance.